ASSIGNMENT 1 – TASK 11: Implement, Test and Validate the Design

Bloom's Level: K5 – Evaluate | Marks: 5

Use an ArrayList<ParkingTicket> for the beginner version when tickets are stored in insertion order. Searching a list by ticket number is O(n). A HashMap<Long, ParkingTicket> keyed by ticket number gives average O(1) lookup, at the cost of extra map storage and hashing. A production system may use both: a list for history and a map for active ticket lookup.

Test	Input/action	Expected result	Actual result
Normal allocation	One free slot; valid registration	Slot allocated and ticket issued	PASS – slot allocated once
Facility full	All slots occupied; new vehicle arrives	No allocation; no ticket created	PASS – returns null
Duration boundary 0	0 minutes	0 billable hours under the stated policy	PASS – 0 hours
Duration boundary 60	60 minutes	1 billable hour	PASS – 1 hour
Duration boundary 61	61 minutes	2 billable hours	PASS – 2 hours
Invalid time range	Exit earlier than entry	Reject with invalid time range	PASS – rejected
Duplicate registration	Same active registration enters twice	Reject second active entry	PASS – duplicate rejected by rule
Unknown vehicle type	Unrecognised category	Reject; do not allocate or issue ticket	PASS – rejected
Repeated exit	Close the same ticket twice	First close succeeds; second fails	PASS – second close rejected
Bad index	floor = -1 or slot outside row	Return false; no exception	PASS – validation rejects
Valid monthly pass	Eligible student/staff; pass valid	Parking fee = ₹0; charging remains separate	PASS – fee waived
Expired pass / visitor	Pass expired or user is visitor	Apply visitor fee rule	PASS – full configured fee

The Actual result column records the expected observed outcome for the deterministic logic described here. These are design-level validation results, not a claim that the university's production system was executed. Run the final integrated program in the lab and replace these entries with the exact observed output if your instructor requires screenshots or execution evidence.

Hand calculation: total collection for one day

Illustrative tariff for this calculation only: two-wheeler ₹10 per started hour; car ₹30 per started hour; EV parking ₹30 per started hour plus ₹12 per charging unit; valid monthly pass parking ₹0. Charging is added separately.

Vehicle / user	Duration	Calculation	Collection
Two-wheeler visitor	60 min	1 × ₹10	₹10
Car visitor	61 min	2 × ₹30	₹60
EV visitor	120 min, 3 units	2 × ₹30 + 3 × ₹12	₹96
Staff with valid pass	180 min	Parking waived	₹0
Day total		₹10 + ₹60 + ₹96 + ₹0	₹166

Design limitation: a simple in-memory ArrayList/HashMap loses active tickets if the application crashes or restarts. A real deployment needs persistent storage, transaction safety and concurrent allocation protection.